

The empirical recurrence for OEIS A301494, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A301494 records “Empirical recurrence of order 77” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 109$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 82$. Its last coefficient is $c_{77} = 1 \neq 0$, so no further tail satisfies anything shorter; any order-77 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A301494 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 1, 3, 4, 5 or 6 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

5, 3, 15, 83, 329, 1576, 7454, 35118, ...

The entry states:

Empirical recurrence of order 77 (see link above)

As of the “Last modified” line on the live entry (Mar 22 2018 11:55:31, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 109$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 109$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 77: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 77.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 218$ exact terms from $n = 5$ on, it returns order 77 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{77} c_i a(n-i),$$

c_1	2	c_{21}	91302	c_{41}	53673	c_{61}	−66400
c_2	9	c_{22}	−46573	c_{42}	−853606	c_{62}	−22841
c_3	15	c_{23}	−262828	c_{43}	−209906	c_{63}	−21994
c_4	21	c_{24}	−372003	c_{44}	−297426	c_{64}	−16722
c_5	−42	c_{25}	−114491	c_{45}	5207	c_{65}	9323
c_6	−46	c_{26}	307233	c_{46}	−69992	c_{66}	−9349
c_7	127	c_{27}	522846	c_{47}	−35175	c_{67}	4584
c_8	331	c_{28}	−37589	c_{48}	424554	c_{68}	−884
c_9	171	c_{29}	−765999	c_{49}	−119721	c_{69}	297
c_{10}	−1499	c_{30}	−1074354	c_{50}	−171369	c_{70}	1041
c_{11}	−4392	c_{31}	−522357	c_{51}	−103674	c_{71}	−153
c_{12}	−6656	c_{32}	581136	c_{52}	103928	c_{72}	−69
c_{13}	−4269	c_{33}	944000	c_{53}	288763	c_{73}	164
c_{14}	−388	c_{34}	828498	c_{54}	306527	c_{74}	−56
c_{15}	6433	c_{35}	−250871	c_{55}	233769	c_{75}	24
c_{16}	15956	c_{36}	−879316	c_{56}	119655	c_{76}	−9
c_{17}	29770	c_{37}	−687410	c_{57}	18653	c_{77}	1
c_{18}	62421	c_{38}	315199	c_{58}	−53889		
c_{19}	99887	c_{39}	1489611	c_{59}	−64531		
c_{20}	137589	c_{40}	729589	c_{60}	−20467		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 82$, and it is the recurrence OEIS A301494 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{77} - c_1 x^{77-1} - \dots - c_{77}$. Its constant term is $-c_{77} = -1$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 77 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 77, so $\deg q = 77$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 77, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 23 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 77, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-77 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A301494.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.